Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets

MPhys Project Report

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Abstract

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1. INTRODUCTION

Commercial nuclear fusion is predominantly pursued via the deuterium-tritium (D-T) fuel cycle due to its relatively large cross-section at achievable plasma temperatures. This reaction fuses deuterium and tritium to yield an alpha particle and a highly energetic fast neutron ($^2\mathrm{H} + {}^3\mathrm{H} \longrightarrow {}^4\mathrm{He} + \mathrm{n}$, $Q = +17.6$ MeV) (Heckrotte and Hiskes, 1971). A breeder blanket is a critical modular component wrapped around the fusion reactor core engineered to fulfil two primary physical functions: capturing and extracting the immense kinetic energy deposited by these 14.1 MeV neutrons for electricity generation, and breeding the requisite tritium to sustain the reactor (Federici et al., 2019; Sawan and Abdou, 2006).

2. THEORY

2.1. Breeder Blanket Physics

2.1.1. Operating Principles

As established in previous foundational work (Khan, 2026a) and Section 1, the viability of commercial deuterium-tritium (D-T) fusion is fundamentally contingent upon net tritium production. Because the D-T plasma yields exactly one neutron per consumption event, and realistic reactor geometries suffer from unavoidable neutron leakage and parasitic absorption, the breeder blanket must achieve a TBR strictly greater than unity to overcome global tritium supply shortages and maintain a continuous fuel cycle (Pearson, Antoniazzi, and Nuttall, 2018). Furthermore, because conservation of momentum dictates that this single emitted neutron carries away four-fifths of the total fusion energy ($E_n \approx 14.1$ MeV compared to $\sim 1 - 2$ MeV neutrons in fission reactors (Heckrotte and Hiskes, 1971)), fusion breeder blankets must be specialised to utilise these ultra high energy neutrons.

To breed this fuel, the blanket exploits neutron capture interactions with lithium. Natural lithium consists of two isotopes, both of which undergo tritium-producing reactions. The dominant ^{6}Li reaction is exothermic and possesses a $1/v$ cross-section dependence, making it highly efficient at capturing thermalised neutrons ($^6\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + 4.8\,\mathrm{MeV}$). Conversely, the ^{7}Li reaction is an endothermic threshold reaction requiring fast neutrons ($^7\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + \mathrm{n}' - 2.5\,\mathrm{MeV}$).

Because the dominant ^{6}Li reaction is purely absorptive, the strict 1-to-1 neutron economy necessitates the inclusion of primary neutron multipliers to maintain a viable population. Heavy elements such as lead are utilised for their substantial spallation cross-sections at high incident energies, undergoing threshold $(\mathrm{n}, 2\,\mathrm{n})$ reactions that effectively double the available neutrons per interaction ($^{208}\mathrm{Pb}(\mathrm{n}, 2\,\mathrm{n})^{207}\mathrm{Pb}$, $Q = -7.37$ MeV). This compensates for macroscopic breeding inefficiencies (Hernández and Pereslavtsev, 2018).

Finally, to increase the capture probability of the ^{6}Li isotope, the 14.1 MeV fast neutrons emitted by the D-T plasma must be moderated down to thermal energies. This moderation is achieved through successive elastic and inelastic scattering collisions within the blanket medium and, if using water (Khan, 2026a), internal coolants. Crucially, these scattering events are the primary mechanism by which the neutron's kinetic energy is deposited into the blanket as extractable heat. The total thermal yield is further modified by the reaction network's kinematics: exothermic captures enhance local heat deposition, whereas endothermic threshold reactions inherently consume kinetic energy, reducing the net thermal yield. Ultimately, to maximise both tritium breeding and energy utilisation while minimising safety risks from neutron leakage, a totally attenuated neutron flux is ideal.

2.1.2. Trace Reactions

The continuous-energy nuclear data libraries utilised by OpenMC resolve interactions across all constituent isotopes, sampled from Brown et al., 2018. Consequently, structural and coolant components exhibit trace background reactions that become visible in high-resolution mesh tallies. Incident fast neutrons ($E_n \approx 14.1$ MeV) trigger minor endothermic spallation events in the structural materials, notably within the zirconium first wall ($^{90}\mathrm{Zr(n, 2\,n)}^{89}\mathrm{Zr}$, $Q = -12.0$ MeV, $\sigma \approx 0.6$ barns) and the EUROFER97 steel separators ($^{56}\mathrm{Fe(n, 2\,n)}^{55}\mathrm{Fe}$, $Q = -11.2$ MeV, $\sigma \approx 0.4$ barns). In addition to multiplication, this unattenuated high-energy flux induces negligible trace tritium production within the first wall via an extreme threshold capture reaction ($^{90}\mathrm{Zr(n, t)}^{88}\mathrm{Y}$, $Q = -11.3$ MeV, $\sigma \sim 1$ µb.). Conversely, highly thermalised neutrons within the coolant layers can undergo radiative capture with the naturally occurring deuterium in water to breed trace amounts of tritium ($^{2}\mathrm{H(n, \gamma)}^{3}\mathrm{H}$, $Q = +6.26$ MeV, thermal $\sigma \approx 0.5$ mb).

2.2. Neutron Transport Metrics

2.2.1. Neutron Track Length

To quantify the neutron population within specific geometric regions of the blanket, stochastic Monte Carlo transport codes, such as OpenMC (Romano et al., 2015), utilise a track-length estimator. Rather than counting individual neutrons crossing a boundary, the simulation integrates the total distance traversed by all particle histories within a defined cell volume (Kalos and Whitlock, 2008).

This yields the volume-integrated scalar flux (Φ_V), which is expressed in units of neutron-centimetres ($\mathrm{n} \cdot \mathrm{cm}$) (Kuridan, 2023) and is formally defined as:

$$\Phi_V = \int_V \int_E \int_{4\pi} \psi(\vec{r}, \hat{\Omega}, E)\, d\hat{\Omega}\, dE\, dV = \sum_i W_i \ell_i \tag{1}$$

where W_i is the statistical weight of the neutron and ℓ_i is the path length of its trajectory through the volume V.

A longer total neutron path length implies a longer residence time within the breeder material, which directly increases the probability of nuclear interactions. The macroscopic reaction rate for any process, such as elastic scattering, is simply the product of the region's macroscopic cross-section (Σ_x) and this volume-integrated flux (Kuridan, 2023).

2.2.2. Radiation Damage and Displacements Per Atom

To accurately assess the structural viability of the breeder blanket and first wall, the neutronic simulation tallies radiation-induced material degradation. The standard metric for quantifying this degradation in a crystalline lattice is Displacements Per Atom (DPA), which represents the average number of times an atom is knocked out of its equilibrium lattice site due to radiation bombardment.

When a high-energy fusion neutron collides with a target nucleus, it transfers kinetic energy, creating a Primary Knock-on Atom (PKA). If the transferred energy exceeds the lattice displacement threshold (E_d), the PKA is ejected and can trigger a secondary collision cascade. However, not all the kinetic energy transferred to the PKA contributes to atomic displacements; a significant fraction is lost to inelastic electronic excitation (ionisation) as the PKA travels through the lattice. The specific fraction of the transferred energy available to cause further elastic nuclear collisions and subsequent atomic displacements is tallied as the "Damage Energy" (T_d).

In the standard Norgett-Robinson-Torrens (NRT) displacement model (Norgett, Robinson, and Torrens, 1975), the raw number of displaced atoms (N_d) is directly proportional to this damage energy:

$$N_{d,\mathrm{NRT}} = \frac{0.8T_d}{2E_d} \tag{2}$$

While the NRT model provides a reliable baseline, it often overestimates the final number of stable defects because it does not account for the athermal recombination of atoms that quickly regain their original lattice positions. To correct for this phenomenon, the Athermic Recombination-Corrected (arc-dpa) model is employed (Nordlund et al., 2018). This introduces a surviving defect fraction, $\xi(T_d)$:

$$N_{d,\mathrm{arc}} = \frac{0.8T_d}{2E_d} \cdot \xi(T_d) = \frac{0.8T_d}{2E_d} \left((1-c) \left(\frac{0.8T_d}{2E_d} \right)^b + c \right) \tag{3}$$

where b and c are material-specific constants dependent on the displaced atom type. Values for the displacement energy (E_d) and these constants for critical structural materials, such as iron (dominating the structural separators) and zirconium (in the first wall), are established in the literature (Saha, Devan, and Ganesan, 2019).

Finally, to determine if a material exceeds its irradiation limits over its operational lifetime, the total number of displacements must be normalised into DPA per Full Power Year (FPY). The DPA received by a structural component over a FPY is given by:

$$\mathrm{DPA/FPY} = \frac{N_{d,\mathrm{arc}}}{\left(\frac{\rho V \cdot N_A}{\mu}\right)} \cdot \frac{P}{E_f \cdot e} \cdot (60^2 \cdot 24 \cdot 365) \tag{4}$$

Here, the denominator calculates the total number of target atoms within the specific component using the material density (ρ), volume (V), Avogadro's constant (N_A), and molar mass (μ). This is multiplied by the annual neutron emission rate—derived from the reactor thermal power (P) and the energy released per fusion event (E_f, converted to Joules via the elementary charge e)—scaled to the number of seconds in a calendar year.

2.2.3. Nuclear Heating and Energy Deposition

In addition to radiation damage, the structural and thermal-hydraulic viability of a blanket design is dictated by the spatial distribution of nuclear heating. Energy is deposited into the blanket materials through two primary mechanisms: neutron kinetic energy transfer during elastic and inelastic scattering, and the absorption of secondary gamma rays (γ-photons) produced during nuclear reactions (Kuridan, 2023).

In neutronic simulations, this localised energy deposition is calculated using Kerma (Kinetic Energy Released in Materials) factors. The total nuclear heating rate, $H(\vec{r})$, at a specific spatial location is the integral of the local scalar flux $\Phi(\vec{r}, E)$ folded with the macroscopic Kerma factor $K(E)$ over all energies (Kuridan, 2023):

$$H(\vec{r}) = \int_E \Phi(\vec{r}, E) K(E)\, dE \tag{5}$$

where $K(E)$ accounts for the sum of all microscopic energy-releasing reaction cross-sections multiplied by the average energy released per reaction. Accurately mapping this heating distribution is critical, as it determines the required mass flow rates and spatial distribution of the coolant layers to prevent material melting and ensure efficient thermal energy extraction for the power cycle. Advanced Monte Carlo codes, such as OpenMC, directly implement these integrals via specialised `heating` tallies to provide high-fidelity energy deposition profiles (Romano et al., 2015).

2.3. ML: Bayesian Optimisation

Given the significant computational expense of evaluating fusion simulations via stochastic Monte Carlo transport, standard grid-search or gradient-descent optimisation methods become computationally intractable across high-dimensional parameter spaces. To navigate this, the study employs Bayesian Optimisation (BO), a sample-efficient global optimisation strategy highly suited for evaluating expensive, "black-box" objective functions (Shahriari et al., 2016).

BO is fundamentally driven by Bayes' Theorem, wherein a "prior" belief about the objective function's landscape is iteratively updated with new simulation evidence to form a refined "posterior" distribution. This loop relies on two coupled mathematical components: a probabilistic surrogate model and an acquisition function.

2.3.1. Surrogate Modelling via Gaussian Process Regression

The surrogate model acts as a computationally inexpensive emulator for the OpenMC physics solver. This project utilises Gaussian Process Regression (GPR) (Rasmussen and Williams, 2006). Unlike parametric regression models that fit a single discrete curve, a Gaussian Process (GP) defines a non-parametric probability distribution over all possible functions that could fit the observed data.

For any given coordinate in the parameter space (e.g., a specific set of blanket layer thicknesses), the GPR outputs a probability distribution characterised by two key metrics:

- **Mean Function, $\mu(x)$:** The expected value of the objective metric (e.g., TBR).
- **Variance, $\sigma^2(x)$:** The uncertainty associated with that prediction.

At locations near previously simulated data points, $\sigma(x)$ collapses to the stochastic noise floor, representing high confidence. In unexplored regions of the parameter space, $\sigma(x)$ expands. This behaviour is governed by the GP's covariance function (or kernel). The kernel mathematically dictates how information "bleeds" between spatially adjacent points, assuming that configurations with similar geometric inputs will yield correlated neutronic outputs. GPR is particularly advantageous for Monte Carlo applications as its kernel can explicitly model the inherent statistical noise of stochastic particle tracking, preventing the optimiser from overfitting to statistical anomalies (Snoek, Larochelle, and Adams, 2012).

2.3.2. Acquisition Functions and the Exploration-Exploitation Trade-off

While the GPR provides a probabilistic map of the design space, it does not dictate the search path. This is the role of the Acquisition Function, an analytically defined, cheap-to-evaluate equation that transforms the GP's mean and variance into a single scalar "utility" score. The parameter space coordinate that maximises this utility is selected as the next candidate for OpenMC simulation.

The acquisition function must carefully balance two conflicting objectives: *exploitation* (sampling areas with a high predicted mean, $\mu(x)$, to refine known local maxima) and *exploration* (sampling areas with high uncertainty, $\sigma(x)$, to discover untracked global maxima).

A standard acquisition metric is Expected Improvement (EI), which calculates the probability and magnitude by which a new sample might exceed the current best observation, f^*:

$$EI(x) = (\mu(x) - f^*)\Phi(Z) + \sigma(x)\phi(Z) \tag{6}$$

where Φ and ϕ are the standard normal cumulative distribution and probability density functions, respectively. While highly efficient for convex problems, EI is prone to stagnation in highly multi-modal landscapes (such as the Rastrigin function (Rastrigin, 1974)). Because EI scales with the probability of beating f^*, it can act "greedily" and refuse to explore deep, uncertain "valleys" if the mean prediction is substantially lower than the current peak.

An alternative is the Upper Confidence Bound (UCB) acquisition function:

$$UCB(x) = \mu(x) + \sqrt{\beta} \cdot \sigma(x) \tag{7}$$

Rather than relying on the probability of improvement against a known baseline, UCB evaluates the absolute best-case scenario for any given point. The hyperparameter β acts as an explicit control over the optimiser's optimism. By setting β to a sufficiently high value, the algorithm is forced to aggressively explore highly uncertain regions (Section 4.3.1 includes demonstrations of this).

3. LITERATURE REVIEW

3.1. Breeder Blanket Architectures

Contemporary fusion reactor designs must achieve a Tritium Breeding Ratio (TBR) strictly greater than 1.0 (as defined in Section 1). While the literature explores diverse material paradigms—ranging from solid ceramics in ITER (Federici et al., 2019) and liquid metal concepts in CFETR (Zhuang, Li, Jian, et al., 2019), to highly compact liquid immersion variants in STEP and ARC (Lord et al., 2024; Sorbom et al., 2015)—evaluating these conceptual breeders reveals significant engineering challenges. Practical blanket modules cannot operate as idealised, homogeneous mixtures; they require complex, layered internal cooling channels and structural separators to ensure thermo-mechanical stability.

A critical review of current mature conceptual designs, such as the Water-Cooled Lithium-Lead (WCLL) and Helium-Cooled Pebble Bed (HCPB), demonstrates that coolants and their associated structural containment typically occupy 30%–40% of the total blanket volume (Del Nevo et al., 2019). Furthermore, the distribution of this necessary non-breeding volume is heavily skewed towards the plasma-facing First Wall (FW). The FW endures the most severe environmental stressors in the reactor; demonstration power plant (DEMO) concepts must mitigate surface heat fluxes often exceeding 1.0 MW/m^2 alongside peak neutron wall loadings of $\sim$ 2.0 MW/m^2 (You et al., 2016). To prevent structural failure under these extreme loads, current literature heavily relies on integrating dense networks of cooling micro-channels within Reduced-Activation Ferritic-Martensitic (RAFM) steels, notably EUROFER97 (Stornelli et al., 2024), frequently augmented by sacrificial tungsten armour (You et al., 2016).

These extreme thermal and neutronic loads strictly dictate component lifetime propositions. While ITER Test Blanket Modules are designed to survive experimental campaigns with cumulative radiation damage below 3 dpa, commercial power plants will subject blanket structures to 20–50 dpa per FPY (Federici et al., 2017). This necessitates a modular design approach, with structural replacement anticipated every 2–5 years to prevent catastrophic embrittlement (Boccaccini et al., 2014).

Consequently, the integration of these extensive structural and cooling networks inherently degrades the global neutron economy. Blanket design ceases to be a purely neutronic exercise and becomes a highly constrained multi-physics trade-off. As demonstrated in recent optimisation studies, researchers must continually refine spatial layer configurations to maximise the primary objectives—TBR and high-grade heat extraction—without exceeding the strict material degradation and operational limits imposed by the heterogeneous environment (Qiu, Lu, Boccaccini, et al., 2018).

3.2. Computational Load in Nuclear Neutronics

High-fidelity neutronic evaluation strictly necessitates stochastic Monte Carlo transport methods (Kalos and Whitlock, 2008), using industry standard codes such as OpenMC (Romano et al., 2015). Achieving statistically significant tallies in deeply shielded or structurally intricate regions imposes a severe computational burden. As explicitly quantified in recent fusion power plant studies by Warmer et al., 2024, sufficiently accurate Monte Carlo neutronics simulations routinely require 10^8–10^9 particle histories, directly leading to several thousand CPU-hours on dedicated supercomputing clusters per run. Consequently, executing traditional exhaustive grid-search optimisations across highly dimensional design spaces results in exponential scaling that is functionally intractable. Humphrey et al., 2023 highlights, in their recent evaluation of fusion component design, the immense computational cost of Monte Carlo transport precludes rapid iterative optimisation, strictly limiting traditional parametric sweeps to low-dimensional or heavily homogenised approximations.

3.3. Machine Learning and Surrogate Modelling in Physics

To circumvent the severe computational limitations of high-fidelity transport codes, the broader computational physics community has increasingly adopted surrogate modelling and sequential learning techniques. Methods such as BO and GPR are distinctly suited to resolving this specific bottleneck (Rasmussen and Williams, 2006). Rather than exhaustively evaluating an entire design space, these algorithms intelligently sample a small fraction of the data points required by a traditional grid search. This targeted approach yields a robust understanding of the optimal parameter landscape while drastically reducing the required number of computationally expensive simulations (Emmerich, Giannakoglou, and Naujoks, 2006; Shahriari et al., 2016).

The translation of this machine learning paradigm into nuclear fusion neutronics is a rapidly emerging, cutting-edge frontier. For instance, Mánek et al., 2023 recently established the efficacy of using ML surrogates to efficiently regress the TBR from heavy Monte Carlo datasets. Crucially, the explicit application of BO to optimise complex breeder blanket architectures is just now being proven as a legitimate, highly competitive route. Concurrent, state-of-the-art work by Humphrey, Brooks, Mungale, et al., 2026 at the UK atomic energy authority (UKAEA) demonstrates the viability of deploying Bayesian sequential learning to navigate the parameter space of an OpenMC neutronic model. The recent emergence of such literature underlines a paradigm shift and validates the timeliness, originality, and necessity of the methodology developed in this study.

4. METHODOLOGY

4.1. Simulation Pipeline Summary

The neutronic performance of the breeder blanket geometries is evaluated using a coupled computational workflow. The 3D parametric CAD models of the tokamak reactor are generated using the Python-based framework Paramak (Shimwell et al., 2021). To ensure computational efficiency while maintaining physical accuracy, the simulation domain comprises a 90° toroidal sector with periodic boundary conditions applied to the bounding surfaces. These Constructive Solid Geometry (CSG) geometries are exported as Direct Accelerated Geometry Monte Carlo (DAGMC) unstructured meshes, which allow for efficient ray-tracing directly on the CAD surface representations without the need for voxelisation. Stochastic neutron transport simulations are subsequently executed upon these meshes using OpenMC (Romano et al., 2015). To ensure statistical robustness and minimise relative error below 1%, simulations utilised high particle counts ranging between 10^5 and 10^7 neutrons per run.

Based on the optimal configurations identified in the previous semester, the material definitions for this architecture are explicitly fixed: the first wall utilises a zirconium alloy, the breeder is a lithium-lead eutectic ($Li_{17}Pb_{83}$), and the coolant is pressurised water (modelled at 600 K and 15.5 MPa). EUROFER97 was selected for the internal structural separators, rear wall, and central solenoid, reflecting industry standards for robust, reduced-activation ferritic-martensitic steels. Furthermore, to more accurately evaluate the spatial distribution of interactions within this refined geometry, the 14.1 MeV Muir-distributed ring source utilised in the previous study has been updated to a default volumetric body source with density decreasing from the major radius centre Romano et al., 2015.

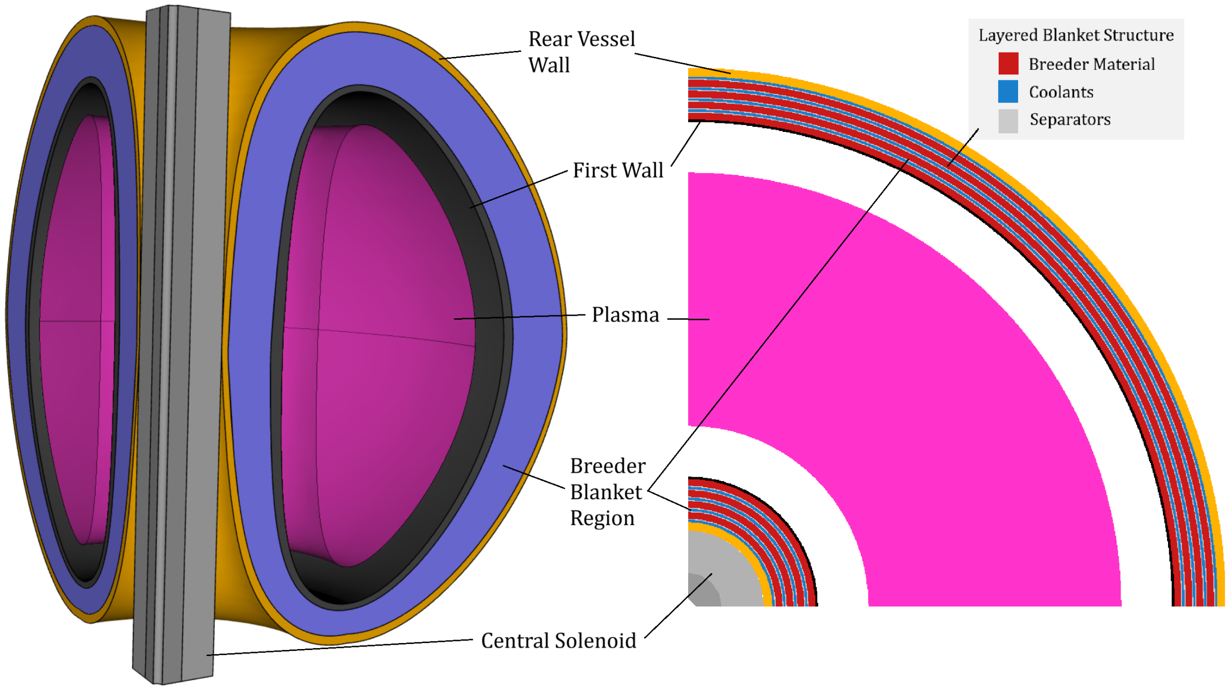


Figure 1: Representative geometry of the simplified Tokamak model generated via Paramak. The simulation domain comprises a 90° revolution with periodic boundary conditions. Included is an example breakdown of the layered blanket, only the coolant and breeder thicknesses are adjusted in this study.

A comprehensive exposition of this foundational workflow was established in the first stage of this study (Khan, 2026a). Building upon those established homogeneous baseline models, this current study transitions to a discrete, heterogeneous layered blanket architecture. We use 4 breeder layers and 4 coolant layers, with separators between each breeder-coolant interface (see Figure 1). This addresses the core unrealistic homogenous approximation of that study, establishing a key step towards more realistic breeder blanket module requirements and challenges. Note that other studies have simulated detailed models with specific coolant piping interlaced in the blanket (see Section 3.1). Paramak's parameterised framework limits the level of heterogeneity to systems with azimuthal symmetry (such as our layered model). The layered model is still an approximation compared to specific real implementations. We proceed as there are key advantages to this approach:

- **Conservative heterogeneity:** Applying this model allows for the majority of heterogeneous effects to be modelled. Importantly, in real implementations streaming effects allow some neutrons to bypass the interactions between coolants and separators. The layered model prevents this, forcing heterogenous material paths. Given heterogeneity negatively affects performance (see Figure 5a), the layered model notably becomes a conservative estimate of blanket performance. In an immature field where specific designs are undecided, a conservative study can be informative.

- **Parametrisation:** A layered model is easily parameterised into a set of thickness variables for each layer. Comprehensively exploring and optimising even this 8-layer model required complex techniques that became a majority component of this project (see Section 4.3). Using a more detailed model necessitates a greater number of parameters and thus exploring higher dimensional spaces. This is non-trivial to optimise, it is instructional to begin with a lower dimensional parameterised model before scaling.

4.2. Neutronic Analysis

4.2.1. Mechanistic Analysis of the Layered Blanket

Initial parametric thickness sweeps were conducted to establish a baseline for the heterogeneous architecture. We also built upon explorations in the first stage of this project (Khan, 2026a) by including tallying of individual layers to understand which were contributing and impactful to global results (see Section 5.1).

Building on this further, mechanistic understanding of performance necessitates localised analysis of several neutron transport parameters throughout the reactor. To generate two-dimensional spatial distributions of nuclear heating and radiation damage, regular Cartesian mesh tallies were overlaid onto the geometry of the blanket model.

In OpenMC, a high-resolution 2D bounding box mesh was defined across the poloidal-radial plane of the blanket segment. Within this mesh grid, six specific reaction metrics were scored independently:

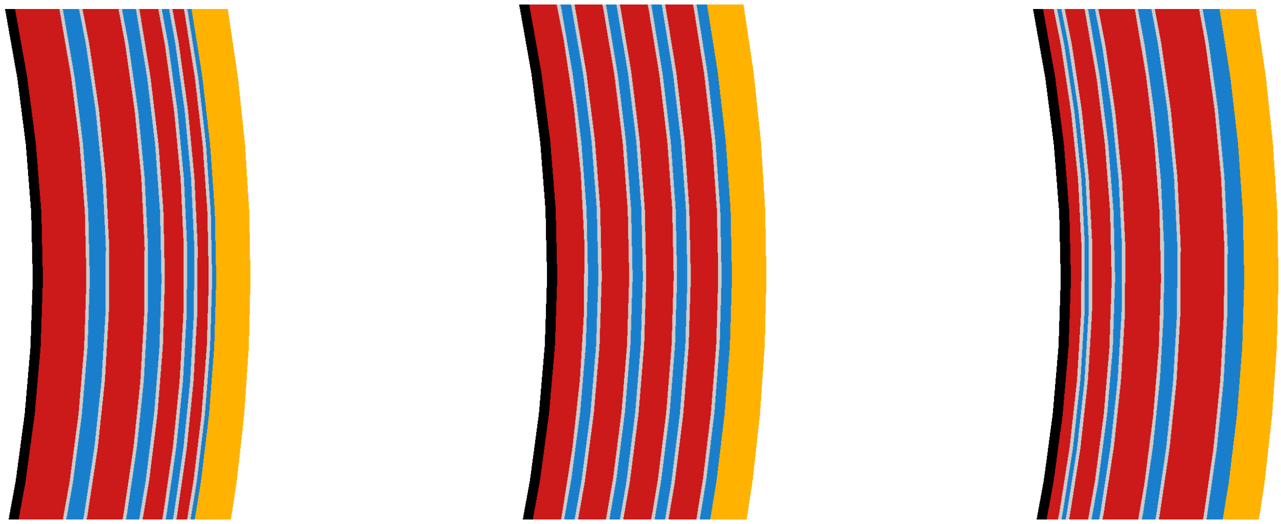
- **Neutron Flux:** Scored using the `flux` tally, measuring the volume-integrated track length of all neutrons across the mesh to map the spatial attenuation of the neutron population.
- **Scattering Density:** Scored using the `scatter` tally, mapping the density of elastic and inelastic collision events primarily responsible for neutron moderation.
- **Tritium Production:** Scored using the `H3-production` tally, identifying the specific geometric regions within the breeder blanket that contribute most heavily to the global TBR.
- **Neutron Multiplication:** Scored using the `multiply` tally, highlighting the precise locations of high-energy threshold reactions (such as $(n, 2n)$ in the multipliers and structural separators) that boost the neutron economy.
- **Nuclear Heating:** Scored using the standard `heating` tally, which records the total kinetic energy deposited by both neutrons and secondary photons per source particle.
- **Damage Energy:** Scored using the `damage-energy` tally, capturing the specific fraction of kinetic energy available to cause atomic displacements in the structural lattice.

The raw outputs from these mesh tallies were subsequently processed and visualised as heat maps (see Section 5.1).

4.2.2. Manual Thickness Distribution Analysis

To systematically investigate the effect of radial material distribution within the heterogeneous blanket, a constrained parametric approach was initially required. The full layered architecture contains eight independent geometric variables (comprising four discrete breeder layers and four coolant layers). Exhaustively exploring this 8-dimensional design space via traditional manual or grid-search methods is computationally intractable.

To reduce the dimensionality of the problem while maintaining physical relevance, a linear gradient parameter, ζ, was introduced as a heuristic constraint (see Figure 2). This parameter governs the relative thickness of the layers across the blanket's depth, providing a human-intuitive method to observe spatial trends. In Section 5.1 we aggregate how neutronic parameters vary under the sweep of ζ, bound to $|\zeta| < 0.35$ from meshing constraints in layers that are too thin.



(a) Forward-weighted ($\zeta < 0$)

(b) Uniform/Baseline ($\zeta = 0$)

(c) Rear-weighted ($\zeta > 0$)

Figure 2: Cross-sectional visualisations of the breeder blanket layers illustrating the effect of the linear gradient parameter, ζ. This parameter controls the radial distribution of layer thicknesses: (a) a forward-weighted geometry shifting volume towards the first wall, (b) the baseline uniform thickness distribution, and (c) a rear-weighted geometry shifting volume deeper into the blanket.

While this constrained, single-variable approach yields valuable insights into the fundamental physics of the blanket, it inherently restricts the exploration of the broader design space. The intractability of manual parametrisation in identifying true global optima across an unconstrained 8-dimensional parameter space necessitate the use of advanced computational techniques. Consequently, the study transitions to the application of Machine Learning algorithms (see Section 4.3) to navigate this complex parameter space efficiently.

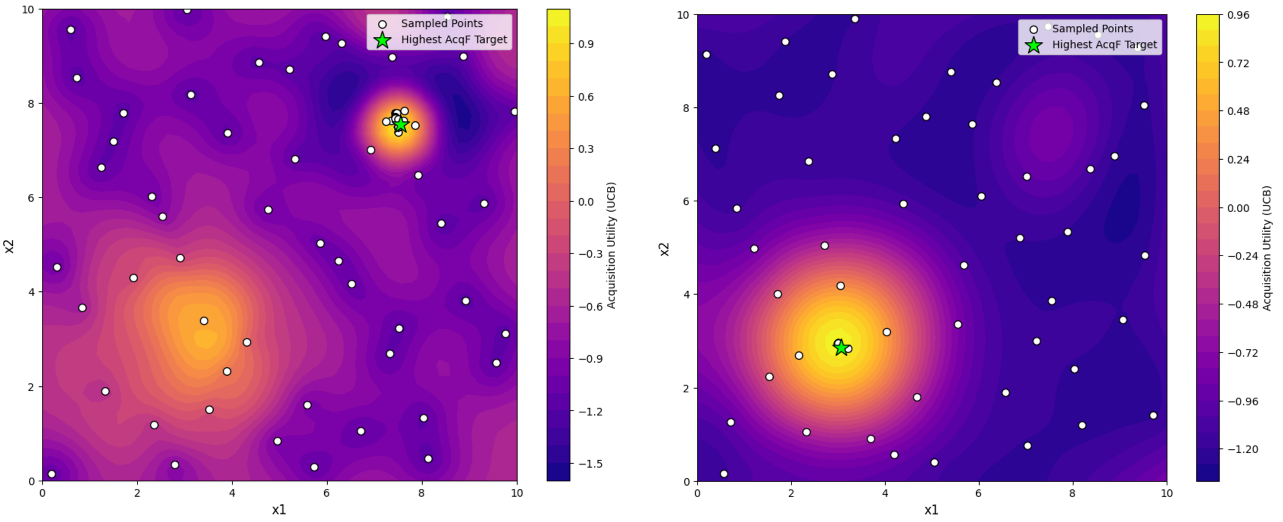
4.3. ML Optimisation

4.3.1. ML Algorithm

To navigate the computationally expensive 8-dimensional design space effectively, Single-Objective Bayesian Optimisation was employed with a Gaussian Process Regressor was utilised as the surrogate model (SOBO-GPR). The optimisation pipeline was developed using the Ax framework, leveraging BoTorch and PyTorch as the underlying computational engines. This scalable, N-dimensional architecture is highly sample-efficient, capable of converging upon optimal configurations within $N < 150$ trials (tested in spaces up to 8 dimensions). This sample efficiency makes it ideal for optimising computationally intensive physical simulations. Furthermore, the objective function and hyperparameters are designed to be modularly configurable, with automated generation of surrogate parameter surfaces and optimisation progress diagnostics.

The objective function was defined by wrapping the discrete parametric OpenMC neutronics simulation into an automated evaluation loop. This function takes the eight blanket layer thicknesses as continuous input parameters (bounded by defined physical engineering constraints) and returns the global TBR as the single scalar objective to be maximised. The computational overhead of the ML solver was negligible (≈ 20 s for $N = 150$ trials); thus, the total runtime was bounded entirely by the neutronics simulation, which required ≈ 60 s per trial on a standard desktop workstation. Because the optimiser was constrained to a maximum of $N = 150$ trials, all ML optimisation studies were completed in under 3 hours. The complete source code developed for this pipeline and complete optimisation results is publicly available (Khan, 2026b).

A critical component of this framework is the selection and tuning of the acquisition function, which dictates how the algorithm samples new data points. For this study, the UCB acquisition function was selected over EI. Unlike EI, UCB provides explicit, scalar control over the exploration-exploitation trade-off via the hyperparameter β. Properly tuning this parameter is vital for a high-dimensional physical system where multiple local maxima may exist. Figure 3 illustrates a 2D projection of the acquisition utility landscape used to benchmark the model's behaviour.



(a) Explorative tuning ($\beta = 1,000,000$)

(b) Greedy tuning ($\beta = 1$)

Figure 3: Comparison of the Upper Confidence Bound (UCB) acquisition function behaviour under different exploration parameters (β). (a) A high β value successfully prioritises exploration, identifying a secondary hidden peak. (b) A low β value results in overly greedy exploitation, causing the model to stagnate on a single local optimum and ignore unexplored regions.

As demonstrated in Figure 3b, setting a low exploration weight ($\beta = 1$) results in a overly greedy search. The algorithm heavily exploits a known local optimum, entirely missing a secondary, hidden peak within the design space. Conversely, Figure 3a shows that a significantly higher parameter ($\beta = 1,000,000$) forces the model to prioritise high-uncertainty regions, successfully mapping the broader landscape and identifying the hidden peak.

Given the complexity and scale of the layered blanket parameter space, premature convergence on a local optimum poses a significant risk. Consequently, the UCB acquisition function was configured with a highly explorative $\beta = 1,000,000$ value to ensure robust, global mapping of the design space before narrowing in on the optimal layer thicknesses

4.3.2. Testing, Tuning & Developing the Optimiser

Before applying the Bayesian optimiser to the high-dimensional neutronic blanket model it was evaluated against a suite of notoriously difficult mathematical landscapes, including the including the Rastrigin (Rastrigin, 1974), Schwefel (Schwefel, 1981), Rosenbrock (Rosenbrock, 1960), and Gramacy & Lee (Gramacy and Lee, 2012) functions, to rigorously benchmark the optimiser's robustness against stagnation. These functions were selected for their diverse pathological traits; such as high-frequency noise, deceptive local minima, and non-symmetric valleys.

These functions were scaled incrementally from 1D up to 4D to ensure the ϵ-greedy and peak-hunting mechanisms remained effective as the search space expanded exponentially.

Figure 4a demonstrates a 1D test case. A critical distinction highlighted here is the difference between the "Actual Best Observed" (the highest discrete point successfully sampled during the simulation) and the "Best Model Prediction" (the theoretical peak calculated by the surrogate model). While finding the absolute optimal TBR is the primary engineering goal, the full continuous surrogate model is arguably the most valuable output for a physicist, providing physical intuition regarding how variables interact across the entire domain.

While 1D tests validate basic functionality, scaling the benchmark to 2D surfaces (Figure 4b, 4c, 4d) is necessary to ensure the algorithm does not fail as dimensionality increases. During these tests, it became apparent that standard Upper Confidence Bound (UCB) acquisition, even with a high exploration parameter, could occasionally stagnate in deep local minima. To prevent this, three bespoke algorithmic enhancements were introduced:

- **Interleaved random injection:** An ϵ-greedy approach was implemented, using a defined probability parameter to periodically force a completely random sample, ensuring exploration in areas the UCB function might otherwise ignore. As the GPR-BO is able to optimise local peaks quickly, we set found setting the probability to $\epsilon = 0.5$ optimal; particularly because we found that UCB acquisition "randomly explored" more in higher dimensions as the space gets exponentially larger.

- **Duplicate avoidance:** Once stuck in a local minimum, the raw BO algorithm would commonly repeatedly test the peak of that minima. Our optimiser was constrained to reject repeated coordinates, forcing continuous spatial exploration and preventing wasted computational resources on identical measurements.

- **Multimodal surrogate exploitation:** Every n_{hunt} trials, the algorithm actively scans the surrogate model to identify the top five predicted peaks. It then forces a sample at the highest peak that remains locally unexplored (highlighted by the pink markers in Figure 4d). In our trials we found $n_{hunt} = 10$ sufficient.

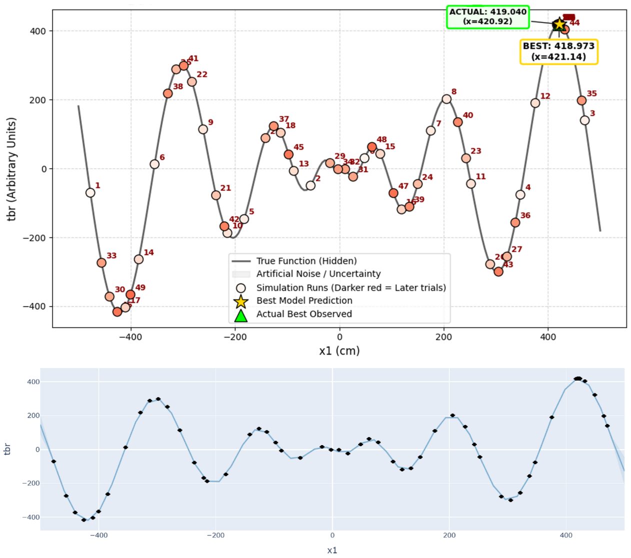
Successfully navigating these notoriously difficult mathematical landscapes demonstrates the optimiser's robust resistance to local minima. This is evidenced by its successful discovery of the 2D hidden peak (Figure 4c) and its highly accurate structural reconstruction of the complex, multi-modal 2D Schwefel function (Figure 4d).

The algorithm was potentially over-prepared for the physical breeder blanket investigation, which is expected to feature broader, continuous global physical trends rather than sharp, discontinuous mathematical spikes. As illustrated by the Gramacy & Lee benchmark (Figure 4b), the model efficiently isolates and solves for broad overall trends, naturally exhibiting less sensitivity to high-frequency, small-length-scale variations. This testing ultimately confirms that while the algorithm maps wider global dynamics highly effectively, microscopic local peaks represent the primary hazard where the model risks entrapment.

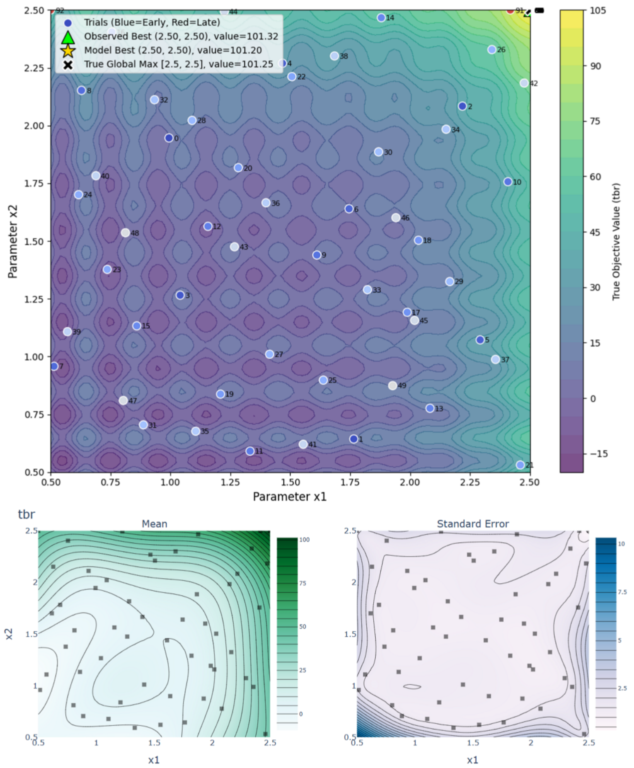
4.3.3. ML Optimisation Studies for TBR-Thickness

To formally bound the Bayesian optimiser against unrealistic values and meshing errors, dimensional limits were imposed on the blanket components based on engineering constraints. Discrete breeder layers were restricted to thicknesses between 3.0 cm and 16.0 cm, while coolant layers were bounded between 1.5 cm and 6.0 cm.

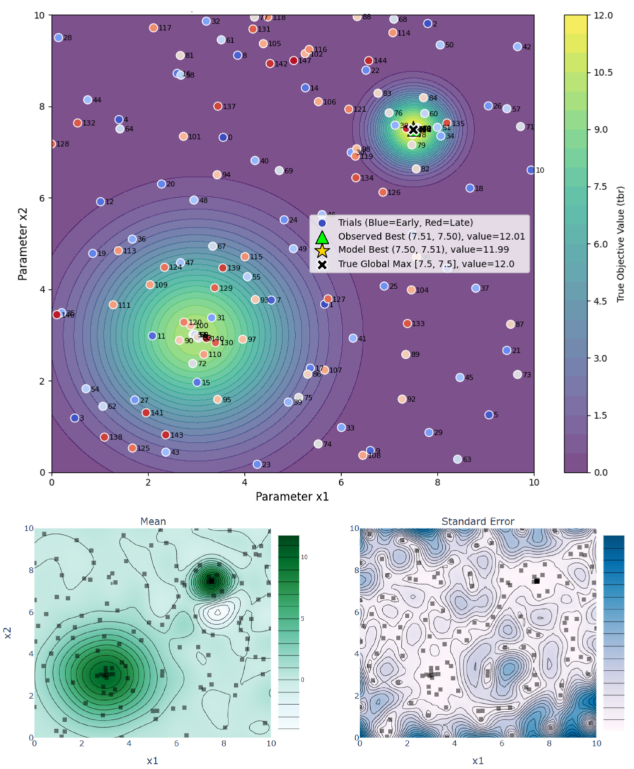
The model was applied on a parameterised version simulation code, in which it had control over the 8 breeder thicknesses (4 breeder layers, 4 coolant layers). It was given a single optimisation objective, to maximise TBR. Throughout testing, monotonic convergence was often



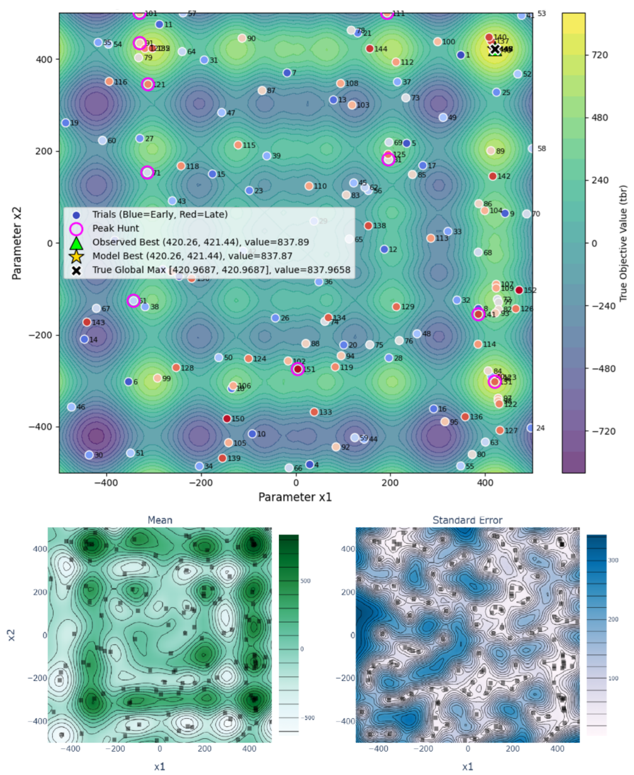
(a) 1D multi-modal Schwefel function test



(b) 2D Gramacy & Lee function test



(c) 2D Hidden Peak function test



(d) 2D multi-modal Schwefel function test

Figure 4: Comprehensive benchmarking of the ML optimisation algorithm across multiple challenging landscapes. (a) A 1D Schwefel test case illustrating full surrogate model reconstruction against the true function. The uncertainty in the model is plotted but too small to be visible. (b), (c) and (d) demonstrate various 2D function tests of the optimiser, where the final mean surrogate model of the space and its paired standard deviation is plotted underneath each. (a) and (b) used $N = 100$ trials whereas (c) and (d) used $N = 150$, where this includes $N_{sobol} = 50$ sobol trials for all. UCB with $\beta = 1,000,000$ was consistent across all trials. Random injection and multi-modal exploitation was implemented for (c) and (d) with $\epsilon = 0.75$ and $n_{hunt} = 10$. (d) includes pink markers to annotate where a secondary "peak hunt" is triggered.

observed (corner/boundary solutions experiencing binding constraints were common) when no additional constraints were applied. Therefore, in the final set of studies we present several optimisation runs in which we applied additional parameter constraints.

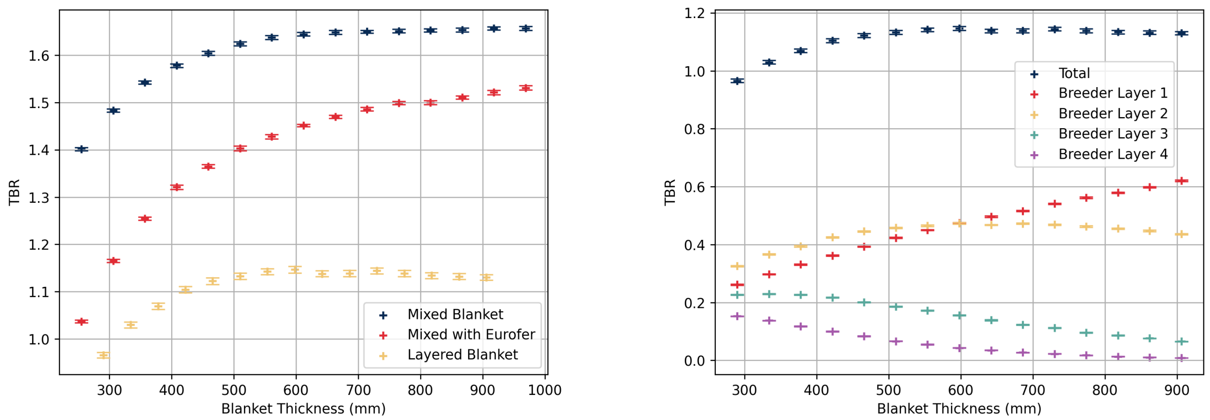
- **8D:** The model is free to adjust all 8 layers, as long as they are bounded within the provided domains.
- **8D Constrained:** A strict total blanket thickness budget of 44.0 cm.
- **4D Breeder Constrained:** Breeder layers were limited to a collective 32.0 cm budget, with all coolant layers locked at their 1.5 cm minimum limit.
- **4D Coolant Constrained:** Coolant layers were required to meet a minimum collective thickness of 12.0 cm, with breeder layers locked at 8.0 cm each.

On receipt of the final ML models and optimised configurations (see Section 5.2), a manual "Binding" configuration was generated to establish a useful comparison for the 8D constrained study. Binding constraints / limits in our model is defined as when the breeder layers are maximised to their 16.0 cm upper bound while minimising the interleaving coolants to 1.5 cm. These prioritise inner layers and continue radially outward until the thickness budget is met.

5. RESULTS

5.1. Neutronic Behaviour of Heterogeneous Architectures

Figure 5a presents the global TBR across these different modelling approaches. The data shows an initial reduction in the calculated TBR when the required structural separator material (Eurofer) is integrated into the homogeneous mixture. A further, more substantial decrease in the global TBR is observed when this exact same material volume is heterogeneously separated into distinct breeder, coolant, and structural layers.



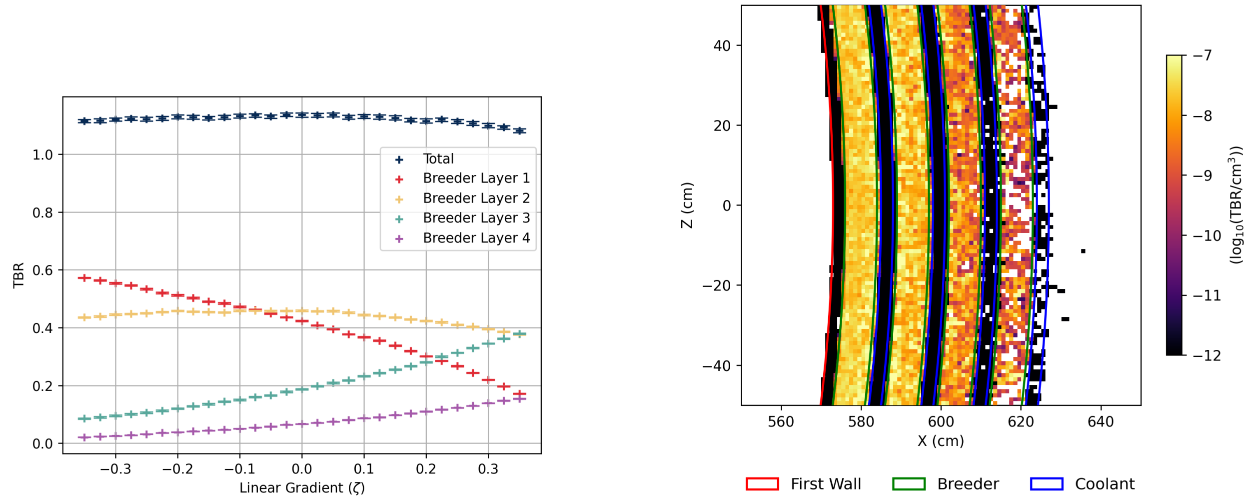
(a) TBR comparison across modelling approximations

(b) Layer-wise TBR contribution vs. thickness

Figure 5: (a) Comparison of TBR against thickness for different blanket models. (b) Demonstrates the layer-wise TBR contributions against thickness

To evaluate layer-wise performance (Figure 5b), breeder and coolant depths were uniformly scaled against constant structural separator thicknesses. Global TBR rapidly saturates at $\approx$ 450 mm. Beyond this threshold, total production remains constant, yet internal breeding shifts: the contribution from Layer 1 rises, Layer 2 stabilises, and deeper layers (3 and 4) contribute progressively less.

The global Tritium Breeding Ratio (TBR) remains relatively constant across the linear gradient parameter (ζ), but the spatial distribution of tritium production shifts significantly between the discrete layers (Figure 6a). The 2D reaction density map (Figure 6b) demonstrates that breeding is asymmetrically focused near the plasma-facing boundary. Notably, at a uniform thickness distribution ($\zeta = 0$), the tritium bred in the second breeder layer actually exceeds that of the first.

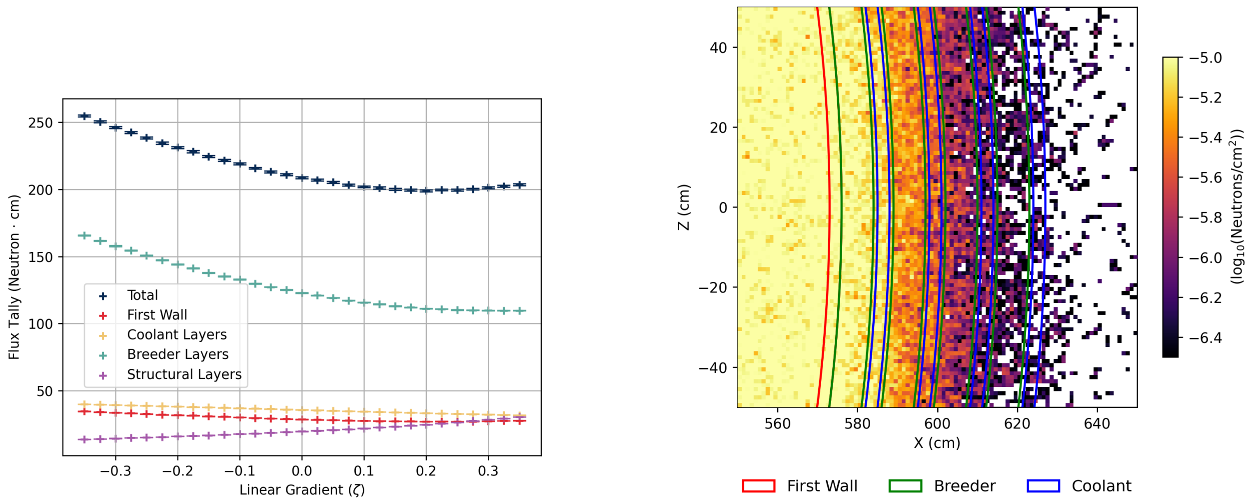


(a) TBR vs ζ

(b) TBR Reaction Density ($\zeta = 0$)

Figure 6: (a) Global and layer-specific TBR performance against ζ, paired with a heatmap of breeding reactions (b).

The volume-integrated flux tally (Figure 7a) indicates that a greater proportion of neutrons survive longer and penetrate further into the blanket in forward-weighted ($\zeta < 0$) configurations. The fluence map (Figure 7b) reveals that the incident neutron flux is completely attenuated by the rear boundary, saturating the blanket. Additionally, a minor step-like attenuation pattern is visible immediately following the discrete coolant layers.

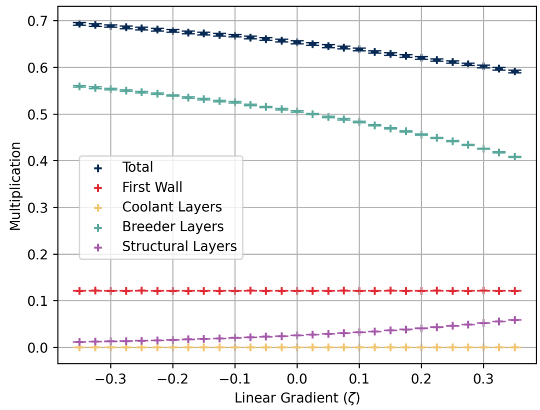


(a) Total Flux vs ζ

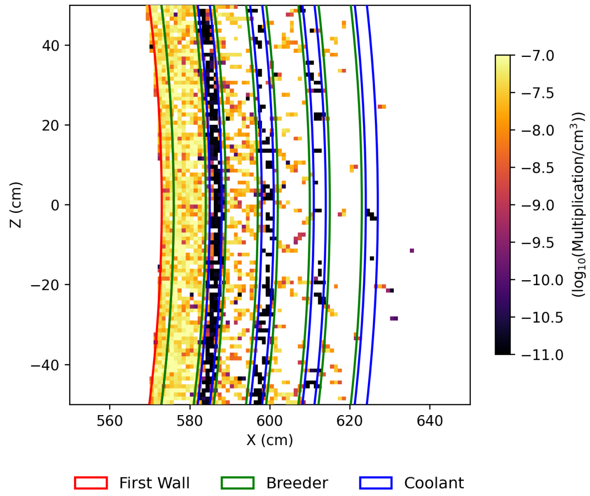
(b) Fluence Map ($\zeta = 0$)

Figure 7: (a) Neutron track length tally within reactor components, paired with a heatmap of neutron fluence depletion through the reactor wall (b).

Figure 8a shows that total neutron multiplication decreases as ζ increases, yielding the highest multiplication rates in forward-weighted configurations. The spatial density map (Figure 8b) highlights that these spallation events are heavily localised within the first two breeder layers.



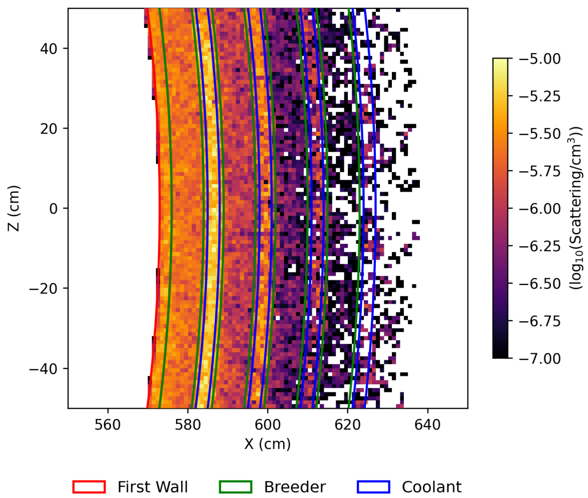
(a) Multiplication vs ζ



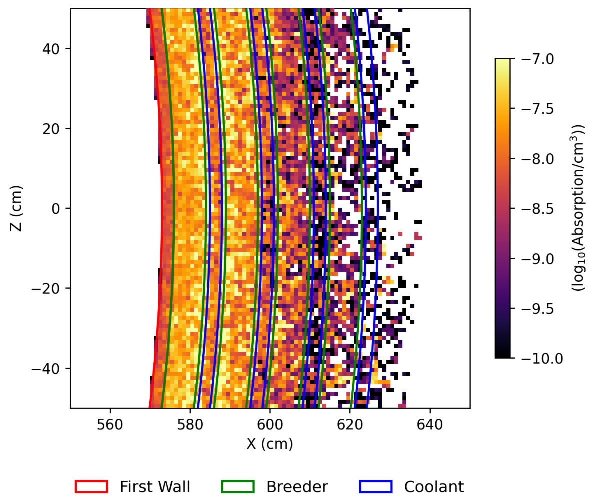
(b) Multiplication Density ($\zeta = 0$)

Figure 8: (a) Total neutron multiplication rate per source neutron split between reactor component, alongside the spatial concentration in the plasma-facing layers (b).

Neutron scattering events exhibit distinct, localised peaks within the water coolant layers (Figure 9a). Correspondingly, Figure 9b shows strong absorption within the breeder material, weak absorption in the first wall, slightly stronger interactions in the coolant, and pronounced, bright absorption peaks within the structural separators. Total absorption remains relatively constant across all ζ configurations, averaging approximately 70% within the breeder, 20% in the separators, and 6% in the coolant (Complete trends over ζ for both scattering and absorption are provided in Appendix A).



(a) Scattering Density ($\zeta = 0$)

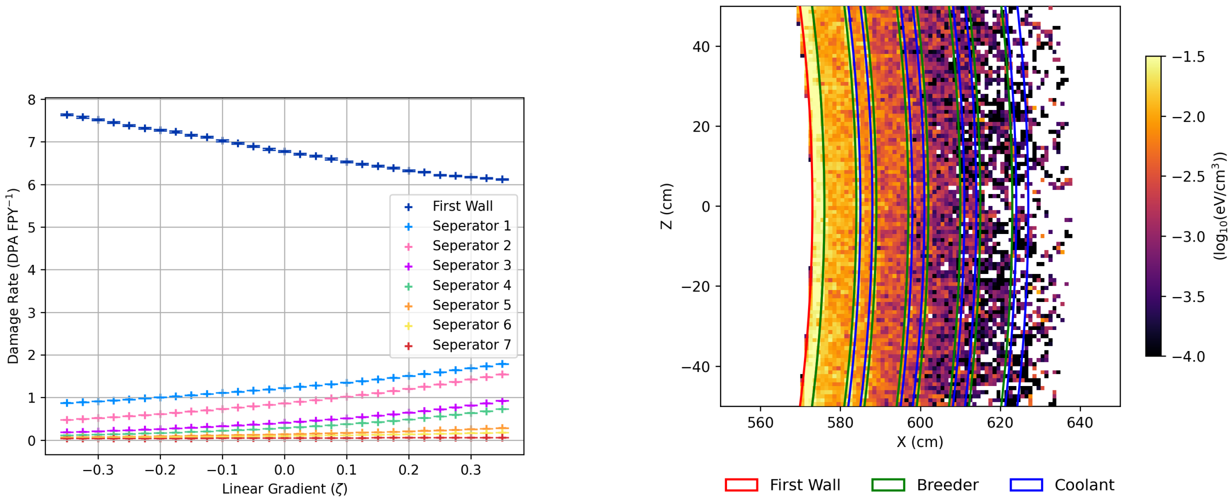


(b) Absorption Density ($\zeta = 0$)

Figure 9: Heat maps of scattering (a) and total absorption interactions (b) in the reactor wall. Absorption tally records all absorptions, including breeding, spallation and parasitic reactions.

Total nuclear heating remains relatively constant at roughly 8 MeV per source neutron across all evaluated geometries (detailed in Appendix A). Of this total energy deposition, approximately 75.0% is deposited in the breeder layers and 19% in the coolant layers. There is greater heating and energy deposition per unit volume in the coolant (heatmap provided in Appendix A).

Finally, Figure 10a tracks structural degradation, showing that the DPA experienced by the first wall increases in forward-weighted configurations ($\zeta < 0$). This coincides with a corresponding increase in total damage energy deposited into the first wall (detailed in Appendix A). The spatial distribution of damage energy (Figure 10b) is heavily concentrated on the extreme plasma-facing layers. Conversely, the DPA sustained by the internal structural separators rises in rear-weighted ($\zeta > 0$) configurations.



(a) DPA Rate vs ζ **(b)** Damage Energy Density ($\zeta = 0$)

Figure 10: Radiation damage metrics across the linear gradient alongside the spatial profile of damage energy deposition.

5.2. ML Optimisation Findings

The optimal blanket configurations identified by the Bayesian optimiser under various dimensional constraints are presented in Figure 11.

The fully unconstrained 8D configuration achieved the highest overall TBR of 1.4055 by pushing all parameters to their binding limits: maximising all breeder layers to 16.0 cm and minimising all coolant layers to 1.5 cm. The partially constrained 4D configurations exhibited lower overall breeding performance. Restricting the optimiser strictly to the maximum breeder budget (TBR: 1.2805) or the minimum coolant budget (TBR: 1.2080) resulted in consecutive decreases in the maximum achievable breeding ratio. Both were similarly under binding conditions, where the 4D coolant study pushed all the coolant thickness as to the outermost layers as much as allowed.

When restricted to the 44.0 cm total thickness budget, the ML model contrastingly identified an “Exponentially decaying” configuration where the breeder layer thicknesses decayed radially outward (16.0 cm, 10.5 cm, 6.9 cm, and 4.6 cm), maintaining minimised coolants, yielding a TBR of 1.3304. By comparison, the manually defined binding constraint configuration yielded an almost statistically identical TBR of 1.3303.

For clarity in evaluating the high-dimensional space, the discrete blanket layers are denoted sequentially from the plasma-facing first wall outwards: breeder layers are labelled B1 through B4, and their adjacent coolant layers are labelled C1 through C4.

To visualise the parameter landscape, 2D projections of the unconstrained 8D surrogate model were generated (Figure 12). These slices plot the thickness of the innermost breeder

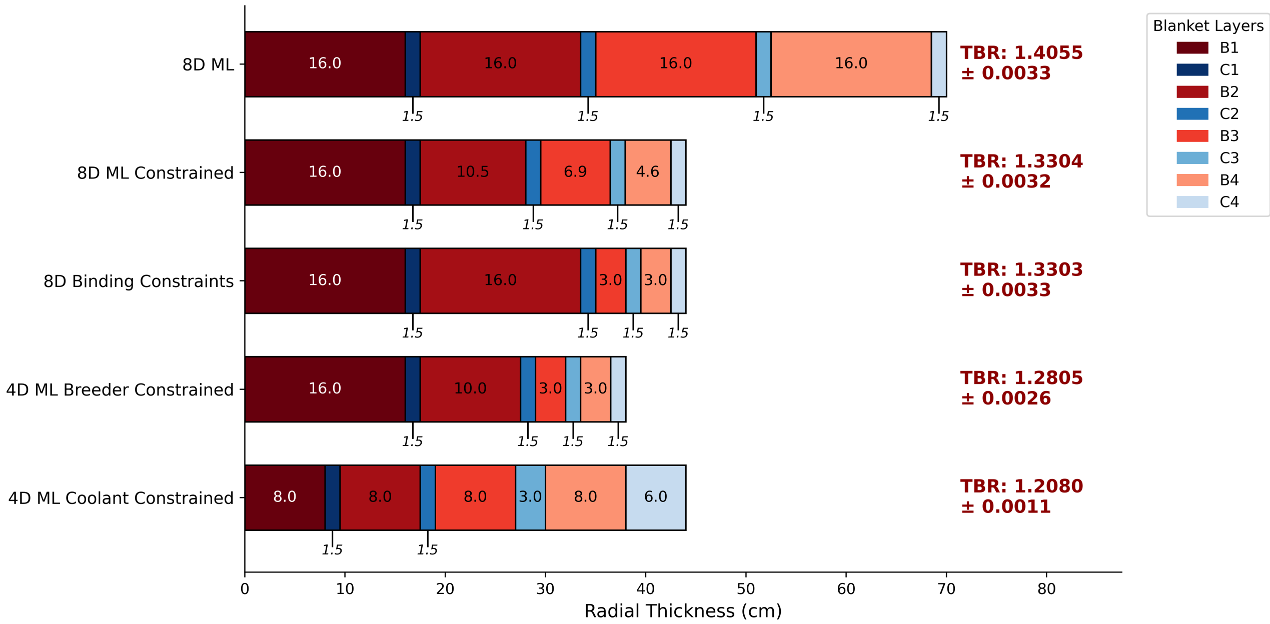


Figure 11: Radial thickness profiles and resulting global TBR for the ML optimised TBR-thickness studies detailed in Section 4.3.3. Associated TBR values are annotated alongside their standard statistical uncertainty. $N = 100$ was used for all except for unconstrained 8D $N = 150$, where these include $N_{sobol} = 50$. Hyper-parameters $\epsilon = 0.5$ and $\beta = 1,000,000$ were tuned for these studies.

layer (B1)—which exhibited the strongest single-variable influence on the Tritium Breeding Ratio (TBR)—against other selected layers.

The gradients within these projections directly illustrate parameter sensitivity. Highly slanted contour lines, such as those mapping B1 against the secondary breeder (B2, Figure 12a) or the inner coolant (C1, Figure 12c), indicate that the TBR is sensitive to variations in both plotted variables. Here the predicted maximum TBR consistently isolates at the boundary corners, favouring maximised breeder thicknesses and minimised coolant thicknesses.

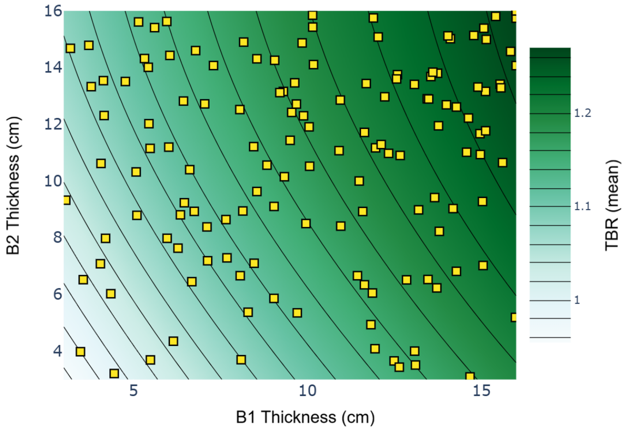
Conversely, projections mapping B1 against the outermost layers, such as B4 (Figure 12b) and C4 (Figure 12d), display nearly vertical contour lines. This indicates that the global TBR is highly insensitive to dimensional changes in the outermost blanket layers compared to the innermost layers.

6. DISCUSSION

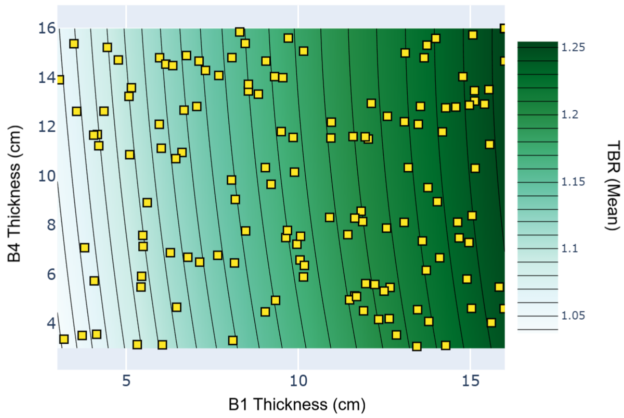
6.1. Physical Implications of Heterogeneous Architectures

Figure 5a demonstrates transitioning from a homogeneous mixture to a discrete layered architecture introduces a distinct neutronic penalty. Geometric separation inherently traps neutrons within isolated water channels and structural separators. Consequently, the plasma-facing breeder region experiences the highest total flux (Figure 7a) but lacks immediate, in-situ moderation. While noted in section 4.1 that this decrease is likely an overestimate, the absorption heatmap (Figure 9b) and Appendix A confirm that these discrete separator (∼20% of absorption) and coolant boundaries (∼6% of absorption) act as pronounced parasitic sinks, driving overall neutron population loss. Furthermore, parasitic absorption within the water coolant is particularly problematic from a safety perspective; neutron capture and cross-contamination can produce tritiated water (HTO), which is approximately 10,000 times more radiotoxic than elemental tritium gas (US Department of Energy, 2015).

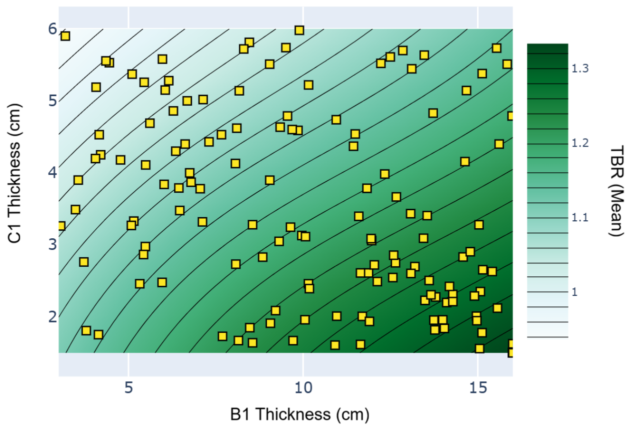
Despite these parasitic losses, the concentrated moderating effect of the water layers (Figure



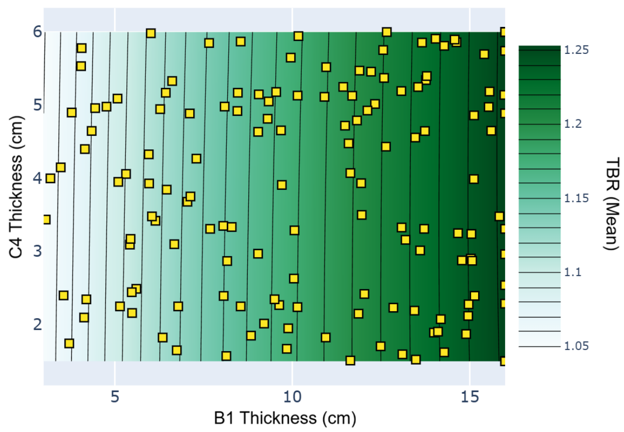
(a) B1 vs. B2 thickness



(b) B1 vs. B4 thickness



(c) B1 vs. C1 thickness



(d) B1 vs. C4 thickness

Figure 12: 2D contour projections of the 8D surrogate model from the unconstrained optimisation run. Each panel plots the thickness of the innermost breeder layer (B1) against a secondary layer (B2, B4, C1, or C4), with all non-plotted dimensions held constant at their optimal boundary limits.

9a) can increase the Li−6 breeding reaction cross-sections in adjacent regions (see Section 2.1.1). As demonstrated in Figures 5b, 6a, and 6b, this thermalisation is so effective that the second breeder layer can actually yield a higher tritium production rate than the first, even though the total available flux has significantly decreased at that depth (Figure 7b).

Across the linear geometric gradient (ζ), the global TBR remains roughly constant, though it begins to marginally decline at higher, rear-weighted ζ values (Figure 6a). This relative stability indicates the blanket is operating in a "saturating condition"; as visually confirmed by the fluence heatmap (Figure 7b), the total radial depth is sufficient to completely consume and attenuate the incident neutron flux regardless of internal distribution.

However, the spatial origin of this breeding is dictated by $(\mathrm{n}, 2\,\mathrm{n})$ multiplication. As detailed in Section 2.1.1, these spallation events require high-energy incident neutrons, causing multiplication to almost entirely deplete after the first two layers (Figure 8b). In rear-weighted configurations ($\zeta > 0$), incident neutrons must pass through structural separators and coolant channels earlier in their transport path. As shown by the scattering density (Figure 9a), this early moderation rapidly degrades neutron kinetic energy, explaining why total multiplication steadily decreases as ζ increases (Figure 8b).

This reduction in high-energy spallation directly links to the varying total flux tallies. Forward-weighted configurations ($\zeta < 0$) sustain a significantly higher total flux (Figure 7a) by generating a larger initial neutron population. However, the modest relative increase in multiplication (roughly $\sim$16.7%) is insufficient to fully account for the massive $\sim$54.5% increase in total flux track length (rising from $\sim$110 to $\sim$170 n·cm). Instead, this increased track length occurs because forward-weighted designs lack thick early water and separator layers to moderate the flux (see Figure 9b and Appendix A for ζ scattering trends). Without early thermalisation, high-energy neutrons simply travel further before being absorbed.

Furthermore, rear-weighted configurations inherently position less water near the high-flux plasma boundary, mitigating early parasitic losses. High-energy neutrons exhibit higher scattering but lower absorption cross-sections in EUROFER97 (Stornelli et al., 2024; Brown et al., 2018). Therefore, rear-weighted designs safely scatter fast neutrons with fewer immediate parasitic losses (see Appendix A). This complex cross-sectional balance explains why forward-weighted designs, despite their vastly superior neutron populations, fail to achieve strictly dominant global TBR.

These geometric shifts also dictate structural viability. In rear-weighted configurations, the internal structural separators are exposed to high-energy source neutrons much earlier, driving a stark increase in internal DPA. Conversely, the first wall experiences peak DPA in forward-weighted configurations, as intense multiplication within the thick first breeder layer causes highly energetic neutrons to backscatter. Given the required DPA-per-year thresholds (yielding a critical embrittlement lifetime of roughly 8.2 to 10.0 FPYs for Zircaloy-4), this concentrated damage highlights the severe material inadequacy of zirconium under intense front-loaded multiplication.

These geometric shifts also dictate structural viability. In rear-weighted configurations, the internal structural separators are exposed to high-energy source neutrons much earlier, driving a stark increase in internal DPA. Conversely, the first wall experiences peak DPA in forward-weighted configurations, as intense multiplication within the thick first breeder layer causes highly energetic neutrons to backscatter. While the absorption heatmap (Figure 9b) demonstrates that zirconium is neutronically superior due to its exceptionally low parasitic absorption, there is a trade-off in its structural threshold. At our calculated damage rates ($\sim$ 6.2–7.5 DPA/year, critical helium embrittlement at 61.99 DPA (Woods, 1966)) Zircaloy-4 restricts the first wall lifetime to roughly 8–10 FPYs—far below ITER's 30-year operational target (Aymar, Barabaschi, and Shimomura, 2002). By contrast, an ultra-fine grained EUROFER97 first wall (2.2–2.5 DPA/year, 57.47 threshold DPA (Woods, 1966)) demonstrates vastly superior structural longevity of 23–26 FPYs, but we found would cause a $\approx$ 9% TBR decrease.

Ultimately, navigating this intractable web of physical variations is practically impossible via manual parameter sweeps. Because proportional scaling of the coolant and breeder layers resulted in largely stagnant global TBR and heating, simple geometric gradients failed to unlock significant performance gains. To break this stagnation and automatically explore the highly-dimensional, non-linear design space, the implementation of a Machine Learning optimisation pipeline became a critical necessity.

7. CONCLUSION

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Note: DOI links are hyperlinked within "[DOI]" placeholder

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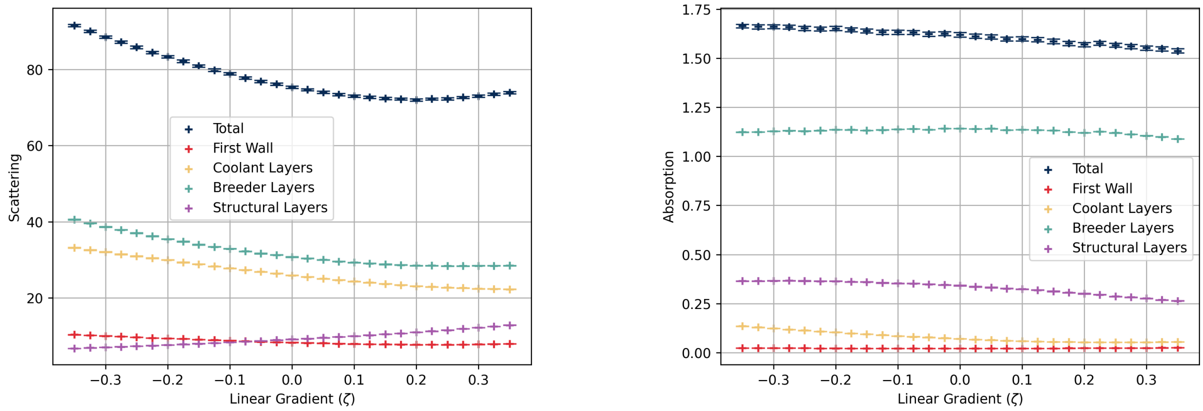
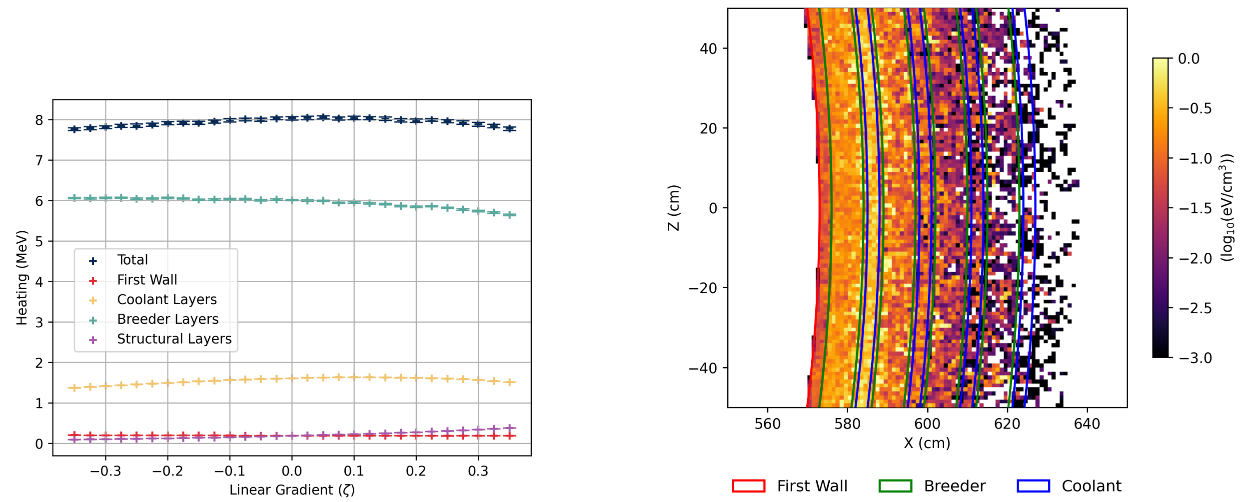
A. SUPPLEMENTARY NEUTRONICS DATA

This section provides the quantitative neutronic trends over the linear gradient parameter (ζ) and supplementary spatial heatmaps referenced in Section 5.1.



(a) Scattering vs ζ **(b)** Absorption vs ζ

Figure 13: Total volume-integrated neutron scattering and absorption events tracked across the geometric gradient.



(a) Heating Energy vs ζ **(b)** Heating Energy Density ($\zeta = 0$)

Figure 14: Total nuclear heating deposition rates and the corresponding spatial concentration within the discrete blanket layers.

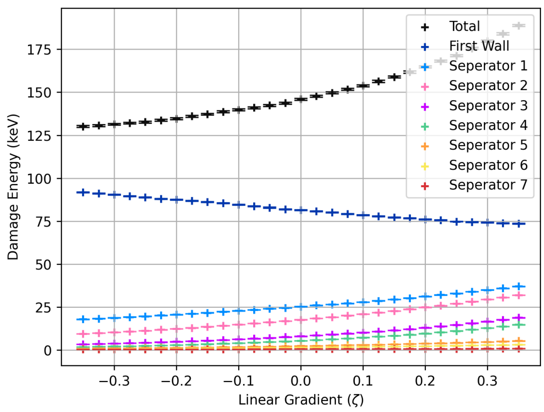


Figure 15: Total damage energy deposited into the plasma-facing first wall and internal structural separators across the geometric gradient.